

App Development Project Report

1. Student Details

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2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and build a web-based application that allows **Admin (Institute), Companies, and Students** to interact with the application based on their roles, such as managing placement drives, tracking applications and recruitments.

Approach:

The app was built using Flask as the backend framework with a modular structure. It allows users to log placement activities, visualize trends through summary charts, and analyze company insights over time.

3. AI/LLM Declaration

I used **GitHub Copilot** to assist in writing SQLAlchemy model definitions.

The extent of AI/LLM usage is around **15–20%**, limited to **code and error fixing suggestions**. All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
SQLite	Lightweight local database for storing user data

5. Database Schema / ER Diagram

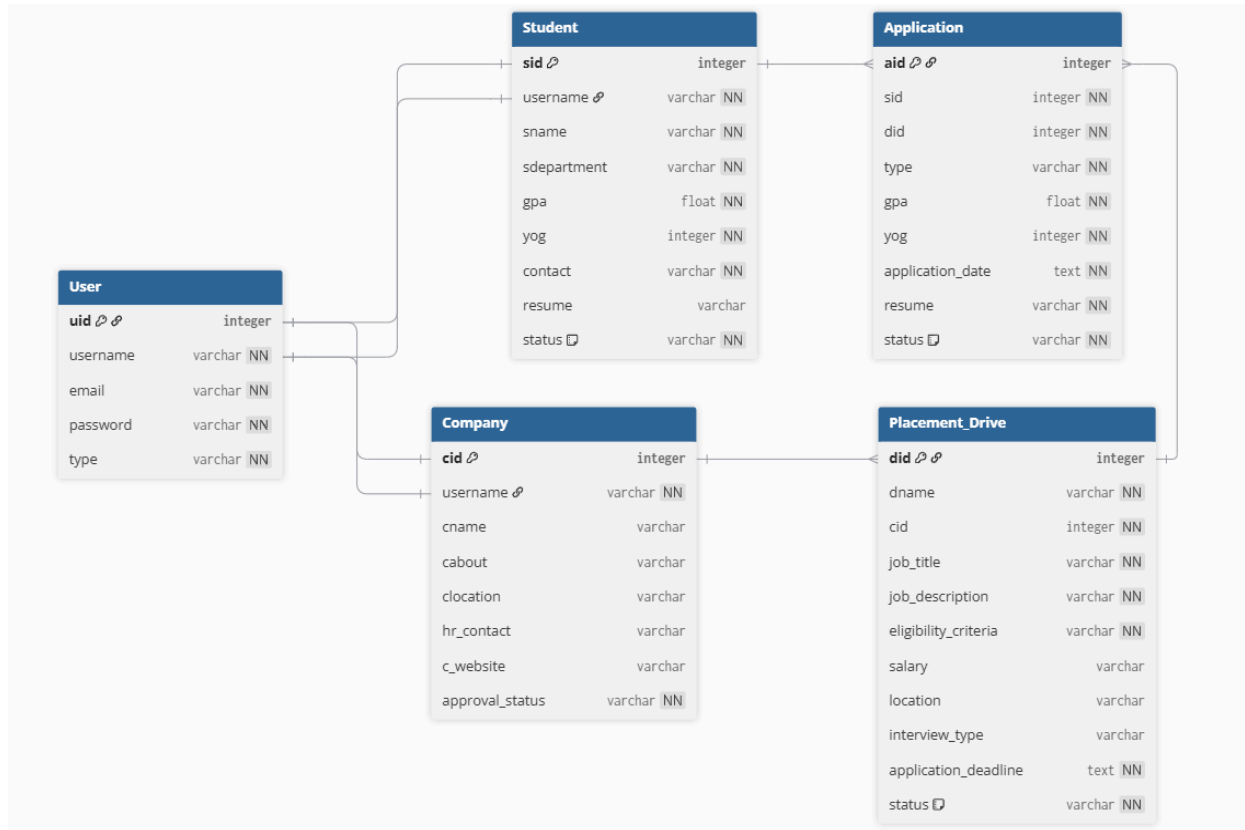
Relations:

1. **User** — stores user profile details (uid, username, email, password, type)
2. **Student** — logs student profile details (sid, username, sname, sdepartment, gpa, yog, contact, resume, status)
3. **Company** — logs company profile details (cid, username, cname, cabout, clocation, hr_contact, c_website, approval_status)
4. **Placement_Drive** — stores drive details (did, dname, cid, job_title, job_description, eligibility_criteria, salary, location, interview_type, application_deadline, status)
5. **Application** — stores student application details (aid, sid, did, type, gpa, yog, application_date, resume, status)

Relationships:

- One-to-One → User → Student
- One-to-One → User → Company
- One-to-Many → Company → Placement_Drive
- One-to-Many → Placement_Drive → Application
- One-to-Many → Student → Application

ER Diagram:



6. Architecture and Features

Architecture Overview:

- **app.py** – main Flask application entry point
- **/models** – database models using SQLAlchemy
- **/database** – to instantiate the database
- **/controllers** – all view functions or routes
- **/templates** – Jinja2 HTML templates
- **/static** – CSS and summary charts images

Implemented Features:

Admin is the pre-existing superuser

Registration for company and students, and login for all (admin, company, student)

Student applications maintenance and placement history logging

No multiple applications by the same student for the same drive

Summary charts

Responsive UI built with Bootstrap

7. Video Presentation

Drive Link:

https://drive.google.com/file/d/1j34ZVepkHrK0gwns44ycrhtiBcl3PiJL/view?usp=drive_link